

CSW + Cisco Secure Firewall Integration Guide

NetFlow (NSEL) visibility and FMC policy enforcement — customer reference

Cisco Secure Workload User Education
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Cisco Secure Workload + Cisco Secure Firewall

NetFlow (NSEL) ingestion and policy enforcement on the firewall

Disclaimer: Companion learning material — not official Cisco product documentation. Validate design, supported versions, and limits against your tenant, release notes, and:

- [FMC Integration Guide \(Cisco.com\)](#)
- [Secure Workload & Firewall deep dive \(secure.cisco.com\)](#)

Architecture diagrams: ARCHITECTURE.md only — do not duplicate diagrams in other docs.

Jorge Quintero — 3-part video series

Cisco Technical Marketing Engineer **Jorge Quintero** ([Cisco Secure Workload YouTube channel](#)) recorded the primary integration walkthrough. Watch **before** implementing.

Part	Title	Topics covered	Watch
1	Secure Workload & Firewall Integration	Introduction, design, high-level architecture, use cases	YouTube
2	Secure Workload & Firewall Integration	Deployment patterns, NSEL ingest, FMC connector setup, policy flow	YouTube
3	Secure Workload & Firewall	Enforcement, telemetry	YouTube

Part	Title	Topics covered	Watch
	Integration	validation, operations, troubleshooting	

Follow-up: [Secure Workload & Secure Firewall Integration Updates \(2025–2026\)](#) — latest behavior on the same channel.

Related Cisco Live session: Jorge Quintero — *Solving the Segmentation Puzzle with Secure Workload* ([BRKSEC-2161](#)).

Executive summary

In hybrid multi-cloud environments, policy controls often fragment across host firewalls, network firewalls, SDN, and cloud security groups — managed by different teams. **Cisco Secure Workload (CSW)** and **Cisco Secure Firewall** together provide a unified microsegmentation path for workloads where **host agents cannot be installed**, while still supporting mixed agent + agentless estates.

Three integrated value streams:

Use case	What it does	Primary path
Microsegmentation	Discover agentless flows (NSEL), model policy (ADM), enforce on FTD via FMC	Secure Firewall Connector + FMC Connector
Virtual Patch	Export workload CVEs to FMC; apply compensating IPS rules until patches ship	FMC Connector (Virtual Patch tab) + agents
Rapid Threat Containment	FMC detects anomaly → Remediation Module → CSW quarantine guardrails	FMC Remediation Module + CSW labels

Solution overview

CSW delivers three platform capabilities that intersect with Secure Firewall:

1. **Zero Trust Microsegmentation** — agent and agentless discovery, ADM-based policy, enforcement on host firewalls, DPUs, **network firewalls**, load balancers, and cloud controls.
2. **Vulnerability Detection and Protection** — runtime CVE visibility from agents; **virtual patch via Secure Firewall IPS**.

3. Behavioral Detection and Protection — process/forensic analytics; **Rapid Threat Containment** with FMC correlation + CSW guardrails.

Architecture

See [ARCHITECTURE.md](#) for all customer-facing diagrams (high-level integration, NSEL/FMC sequences, scope mapping, insertion options, virtual patch, rapid threat containment). This guide covers steps and design detail only — no duplicate diagrams here.

Two connectors — do not mix them

Integration	Connector	Purpose	Data direction
Flow visibility (NetFlow / NSEL)	Secure Firewall Connector on Ingest appliance	CSW sees flows from FTD/ASA without agents	Firewall → Ingest → CSW
Policy enforcement	FMC Connector (Cisco Secure Firewall in UI)	CSW pushes segmentation to FTD via FMC	CSW → FMC → FTD

Also available	Use when
NetFlow connector	Switches/routers export NetFlow v9/IPFIX — not FTD NSEL
Host agents	Process-level visibility + host enforcement
ACI connector	Fabric policy — separate from this FTD/FMC guide

Workload protection level (design first)

Before choosing firewall insertion mode, define the **security/trust boundary** your organization uses (subnet, zone, app tier, persona). The white paper recommends aligning network engineers and app owners on a measurable protection level — commonly **subnet-level** for agentless FTD insertion.

Persona	Typical boundary	Agent vs agentless
App / DevOps	App tier, namespace	Often agent-based
Network / Firewall	Subnet, zone, VRF	Often agentless (this guide)
Cloud	VPC/VNet, security group	Mixed

This drives whether you use **L2 transparent** (fine-grained), **L3 routed** (faster zone segmentation), or **cloud hub** patterns — see [Insertion options](#).

Prerequisites checklist

CSW + Ingest (visibility)

#	Requirement	Notes
1	CSW tenant (SaaS or on-prem) with admin access	Scopes and workspaces ready
2	Secure Workload Ingest virtual appliance	Active before enabling connector
3	Secure Connector (SaaS) or network path	Mandatory for SaaS
4	FTD/ASA can reach Ingest on UDP 4729	Copy IP:port from connector details
5	Capacity	45k fps per connector; 135k fps per Ingest appliance (on-prem)

FMC (enforcement)

#	Requirement	Notes
1	FMC or cdFMC with supported FTD devices	FTD managed by FMC only
2	FMC REST API user with administrative privileges	Required for connector
3	FTD associated with FMC; ACP assigned; deploy verified	Test FMC → FTD deploy before CSW
4	Secure Connector if SaaS CSW cannot reach FMC directly	HTTPS 443 to FMC
5	FMC HA: include standby FMC IP in connector	Auto-switch on failover
6	Supported versions	Segmentation: FMC 7.0.x+ (dynamic objects 7.0.1+); Virtual Patch: 7.2+

Virtual Patch (optional)

- CSW **agents** on target workloads
- Threat intelligence **CVE data pack** on CSW cluster

- Workloads use **FTD as default gateway** (traffic through IPS path)
 - FMC network discovery sees hosts under **Analysis** → **Network Map**
-

Part A — NetFlow / NSEL visibility (Secure Firewall Connector)

Goal: CSW receives bidirectional, stateful flow records from the firewall **without** host agents.

Step A1 — Plan ingest placement

1. **Manage** → **Workloads** → **Virtual Appliances** — confirm Ingest is **Active**.
2. Record **connector slot IP** and port (**4729** default).
3. Open firewall path: **UDP 4729** from FTD/ASA to Ingest.
4. **VRF: Manage** → **Agents** → **Configuration** → **Agent Remote VRF Configurations** — one connector per VRF.

Videos: [Connector Overview](#) · [Connector Deployment](#)

Step A2 — Enable Secure Firewall Connector in CSW

1. **Manage** → **Workloads** → **Connectors**.
2. Select **Cisco Secure Firewall Connector** (formerly ASA connector).
3. **Enable** on **Ingest appliance** (not Edge-only).
4. Record connector **ID** and **IP:port** from connector details page.

Limits: **1** Secure Firewall connector per Ingest appliance; **10** per tenant.

Step A3 — Enable NSEL on Secure Firewall

ASA example (replace <INGEST_CONNECTOR_IP>):

```
flow-export destination outside <INGEST_CONNECTOR_IP> 4729
flow-export template timeout-rate 1
!
policy-map flow_export_policy
  class class-default
    flow-export event-type flow-create destination <INGEST_CONNECTOR_IP>
    flow-export event-type flow-teardown destination
<INGEST_CONNECTOR_IP>
    flow-export event-type flow-denied destination <INGEST_CONNECTOR_IP>
    flow-export event-type flow-update destination <INGEST_CONNECTOR_IP>
    user-statistics accounting
service-policy flow_export_policy global
```

FTD: Configure NSEL export via FMC/device template — [ASA NetFlow Implementation Guide](#).

Videos (Jorge Quintero, 3-part series): [Part 1](#) · [Part 2](#) · [Part 3](#) · [2025–2026 updates](#)

Step A4 — Validate flow ingestion

1. Generate permitted and **denied** test traffic.
2. **Flow Analysis** — confirm firewall-sourced endpoints.
3. Connector heartbeat healthy under **Manage** → **Workloads** → **Connectors**.
4. Optional API filters: `user_src_NETFLOW_IDENTIFIED` / `user_dst_NETFLOW_IDENTIFIED`.

NSEL handling: flow-create/update → exported; flow-denied → **rejected** disposition; connector sends **forward** + **reverse** flows; **NAT flow-stitching** for end-to-end visibility.

Video: [Flow Analysis](#)

Step A5 — Label workloads and run discovery

1. Import **labels** from CMDB (ServiceNow), IPAM, or manual assignment.
2. Run **ADM** on scoped workspace.
3. Remain in **Monitor** mode until app owners validate dependencies.

Videos: [Basic Application Discovery](#) · [Enhancing Application Discovery](#) · [ADM & Policy Analysis](#)

Part B — Policy enforcement on Secure Firewall (FMC Connector)

Goal: CSW pushes **L3/L4 segmentation** to FTD via FMC using **Dynamic Objects** — inventory changes refresh membership without manual object edits.

CSW orchestrated rules are **L3/L4 east-west microsegmentation**. Layer-7 FMC features should not be added to CSW-controlled rules (not supported).

Step B1 — Onboard firewalls in FMC

1. FTD devices registered to **FMC** or **cdFMC**.
2. Each FTD assigned to an **Access Control Policy (ACP)**.
3. Verify FMC can **deploy** to FTD and traffic behaves as expected.
4. Document **one scope** → **one ACP** mapping plan (1:1 only).

Video: [FMC Integration \(Edge / Ingest / Appliance\)](#)

Step B2 — Create FMC Connector in CSW

1. **Manage** → **Workloads** → **Connectors** → **Firewall** → **Cisco Secure Firewall**.
2. Click **Configure your new connector here**.
3. Complete **New Connection** fields:

Field	Description
Connector Name	Unique name for this FMC connection
Username / Password	FMC REST API credentials (admin privileges)

Field	Description
CA Certificate	FMC CA cert for TLS validation, or Disable SSL on trusted networks
Server IP/FQDN and Port	FMC hostname or IP (443)
HTTP Proxy	If required: <proxy.host>:<proxy.port>
Secure Connector	Enable when SaaS CSW reaches FMC through Secure Connector

4. Click **Create**; verify **Event Log** tab for errors.

Official procedure: [FMC Integration Guide — Connection Settings](#)

Step B3 — Map scope to Access Control Policy

1. Open FMC connector → **Segmentation** tab → + **Add**.
2. **Add ACP Mapping:**
 - Select **Access Policy** (ACP) from FMC.
 - Map to exactly **one Scope**.
 - **Use Secure Workload Catch-All** — CSW catch-all in Default section, or use FMC default action.
 - **Enforcement Mode:**
 - **Merge** — CSW rules coexist with FMC rules; configure priority.
 - **Override** — CSW replaces user-created rules in mapped sections.
 - **Absolute / Default Policies priority** (Merge only):
 - Insert **above** or **below** existing Mandatory/Default rules per section.
3. Click **Submit**.

Scope mapping strategy

Model	When	Behavior
Child / leaf scope	Single application behind firewall	Push that app's policies + parent guardrails
Parent scope	Multiple applications	Push parent + eligible child policies (hierarchical)

Topology awareness

CSW pushes rules only to FTD devices on the **traffic path** for the mapped scope — avoids rule sprawl across unrelated firewalls.

Videos: [Policy Enforcement Overview](#) · [Where to Enforce](#) · [Policy Ordering](#)

Step B4 — What CSW creates in FMC

When enforcement is enabled on a workspace, CSW converts policies to FMC rules:

FMC artifact	Prefix	ACP section	Purpose
Golden rules	Workload_golden__	Mandatory	CSW ↔ agent communication behind FW
Segmentation rules	Workload__	Mandatory / Default	Enforced workspace policies
Catch-all rules	Workload_ca__	Default	If Use Secure Workload Catch-All enabled
Dynamic objects	Workload0bj__	Objects	Scope/inventory IP membership

CSW policy type	FMC placement
Absolute	Mandatory category
Default	Default category

Do not manually edit Workload_* rules in FMC — CSW restores them on next push. Independent FMC rules are safe in **Merge** mode if they avoid these prefixes.

Step B5 — Model policy, then enforce

1. Build or accept **ADM-recommended** policies in workspace tied to scope.
2. **Simulate** / **Monitor** — Policy Visual and flow analysis.
3. Change window: **enable enforcement** on workspace.
4. Confirm FMC deployment succeeds; verify Dynamic Objects update as inventory changes.

Videos: [Policy Lifecycle](#) · [Policy Validation and Analysis](#)

Step B6 — Validate enforcement

Test	Expected result
Allowed business flow	Passes; CSW/FMC show permit
Intentionally denied flow	Blocked at FTD; CSW rejected-flow evidence
New workload in scope	Workload0bj__ membership updates
FMC deployment	Automatic on CSW policy change
Compliance monitoring	Alerts on rejected flows / policy deviation

Check **Event Log** on FMC connector for push errors (Information / Warning / Error).

Part C — Virtual Patch (optional)

Compensating **IPS** control for CVEs until permanent patches are deployed.

1. FMC connector → **Virtual Patch** tab → **Create a Virtual Patching Rule**.
2. Define **Host Query** (scope/filter) and **CVE Query** (CVSS, Cisco Risk Score, etc.).
3. CSW exports matching CVEs to FMC **Third-Party Vulnerabilities**.
4. FMC admin runs **Cisco Recommended Rules** to map CVEs → Snort signatures.
5. Apply IPS policy on ACP rule for affected traffic flows.

Prerequisites: Agents on workloads; CVE data pack; FTD as gateway; FMC 7.2+.

Part D — Rapid Threat Containment (optional)

Automated quarantine when FMC detects malicious behavior:

1. Install **FMC Remediation Module for Secure Workload** on FMC.
 2. Configure CSW cluster IP, root scope, API key (**User Data Upload** permission).
 3. Define **Correlation Rules** and **Correlation Policy** responses.
 4. On trigger, Remediation Module sends affected IPs to CSW via API.
 5. CSW **guardrail policies** (quarantine labels) enforce across agents and FTD.
-

Firewall insertion options

Mode	Segmentation	NSEL visibility	Best fit
L2 Transparent	Intra- + inter-subnet	Full	Legacy OS, fine-grained, localized
L3 Routed	Inter-subnet only	Inter-subnet only	Quick zone segment, dev/non-prod
ACI Service Graph PBR	Intra + inter EPG/ESG	Full (FW in path)	ACI + FTD multi-policy
AWS hub VPC	Inter-VPC	VPC logs + NSEL	Centralized cloud east-west
AWS distributed	Intra-VPC	VPC logs + NSEL	Per-VPC enforcement
Azure hub VNet	Intra- + inter-VNet	NSG logs + NSEL	UDR steering
GCP hub VPC	Inter-VPC	VPC logs + NSEL	Centralized GCP

All patterns support **dual management**: CSW east-west + FMC north-south on the same ACP (Merge mode).

FAQs (from Cisco white paper)

Question	Answer
Mixed agent + agentless behind same mapped firewall?	Works. Agent-based rules may also appear in ACP (hierarchical policy). Primary use case remains agentless protection.
Map multiple scopes to one ACP?	No — strictly 1 scope : 1 ACP .
Add Layer-7 / FMC features to CSW rules?	Not supported — CSW rules are L3/L4 east-west only.
Dual-management (CSW + FMC rules)?	Merge mode + rule ordering; preserve order after go-live.
Modify a CSW-owned rule in FMC?	CSW overwrites on next push.
Supported FMC versions?	Dynamic objects: 7.0.1+ ; Virtual Patch: 7.2+ ; qualified: 7.0.1, 7.2.5 .

Recommended video learning path

Order	Video	Presenter / channel	Link
1	Secure Workload & Firewall Integration — Part 1	Jorge Quintero · @ciscosecureworkload	Watch
2	Part 2 — deployment & policy flow	Jorge Quintero	Watch
3	Part 3 — enforcement & operations	Jorge Quintero	Watch
4	Integration Updates (2025–2026)	Cisco Secure Workload channel	Watch
5	Connector Overview	Cisco Secure Workload channel	Watch
6	Connector	Cisco Secure	Watch

Order	Video	Presenter / channel	Link
	Deployment	Workload channel	
7	FMC + Edge / Ingest / Appliance	Cisco Secure Workload channel	Watch
8	Where to Enforce	Cisco Secure Workload channel	Watch
9	Policy Enforcement Overview	Cisco Secure Workload channel	Watch

NetFlow connector vs Secure Firewall connector vs API flowsearch

Method	Use when	Scale notes
Secure Firewall Connector (NSEL)	FTD/ASA agentless visibility	45k fps per connector; 135k fps per Ingest
NetFlow connector	Switches/routers (v9/IPFIX)	~15k fps per connector
Host agents	Process-level + host enforce	Best fidelity
OpenAPI flowsearch	Query ingested flows	Paginate; ≤24h; 1h chunks — not bulk ingest

POV evidence checklist

Evidence	Proves
Connector enabled + heartbeat healthy	Ingest path live
Sample flows in Flow Analysis (NSEL)	Visibility
ADM map for scoped app	Discovery value
FMC connector connected + Event Log clean	Enforcement path
Scope ↔ ACP mapping screenshot	Topology binding
Workload__ / WorkloadObj__ in FMC	Policy push working
Monitor-mode policy stats	Safe pre-enforce review
Post-enforce deny test + FMC hit	Enforcement works

Evidence	Proves
Allowed transaction test	No business break

Common pitfalls

Pitfall	Fix
Confusing NetFlow connector with Secure Firewall connector	NSEL from FTD/ASA → Secure Firewall Connector only
Expecting NSEL alone to enforce	Requires FMC connector + workspace enforce
Editing Workload_* rules in FMC	CSW will overwrite — change policy in CSW
Multiple scopes on one ACP	Redesign — 1:1 mapping only
L3 routed FW for intra-subnet segment	Use L2 transparent or host agents
Bulk flowsearch API timeouts	Paginate; chunk time windows
Skipping Secure Connector on SaaS	Required for Ingest and FMC reachability

Official Cisco documentation

Document	URL
FMC Integration Guide	cisco-secure-workload-and-fmc-integration-guide.html
Deep dive white paper	secure-workload-whitepaper
CSW 4.0 SaaS — Connectors	m-connectors.html
Marketing white paper	sec-workload-firewall-wp.html
ASA NSEL configuration	ASA NetFlow Implementation Guide

Related repos

- [CSW-Secure-Firewall-Integration-Guide](#) — **this repo**
- [CSW-User-Education](#) — video library and onboarding
- [CSW-Policy-Lifecycle](#) — ADM → Monitor → Enforce
- [CSW-Agent-Installation-Guide](#) — host agents
- [CSW-Identity-Integration-Guide](#) — AD / Entra identity
- [CSW-ServiceNow-Connector-Guide](#) — CMDB labels
- [csw-splunk-integration](#) — SIEM
- [CSW-Compliance-Mapping](#) — compliance frameworks
- [csw-logs-check](#) — host enforce timing from agent logs

CSW + Secure Firewall — Architecture Reference

Single source of truth for all architecture diagrams in this repo. Other documents link here instead of repeating diagrams.

Video walkthrough: Cisco TME **Jorge Quintero** — 3-part series on [@ciscosecureworkload](#):
[Part 1 — design](#) · [Part 2 — deployment](#) · [Part 3 — operations](#)

Derived from the [Cisco Secure Workload & Secure Firewall deep dive](#) and the [FMC Integration Guide](#).

Disclaimer: Companion material — validate against your CSW and FMC release notes before production.

1. High-level integration (microsegmentation)

This is the reference architecture for **agentless east-west segmentation** using NSEL visibility and FMC enforcement.

flowchart TB

```
subgraph Sources["Context & telemetry"]
    CMDB["CMDB / ServiceNow"]
    IPAM["IPAM / DNS"]
    Labels["Manual labels"]
end

subgraph Workloads["Application workloads"]
    Agent["Agent-based workloads<br/>host firewall enforce"]
    Agentless["Agentless workloads<br/>legacy / appliance / no agent"]
end

subgraph Visibility["Visibility path – NSEL"]
    FTD["Secure Firewall FTD / ASA<br/>Transparent or Routed mode"]
    NSEL["NSEL export<br/>NetFlow v9 · UDP 4729"]
    Ingest["Secure Workload Ingest Appliance"]
    SFC["Secure Firewall Connector<br/>Docker · up to 45k fps each"]
    FTD -->|"flow-create · teardown · deny · update"| NSEL
    NSEL --> SFC
    SFC --> Ingest
end

subgraph CSW["Cisco Secure Workload"]
    FlowDB["Flow analysis & inventory"]
    ADM["Application Dependency Mapping"]
    Policy["Policy engine<br/>Monitor → Simulate → Enforce"]
    Ingest --> FlowDB
    CMDB --> FlowDB
    IPAM --> FlowDB
    Labels --> FlowDB
end
```

```

    FlowDB --> ADM --> Policy
end

subgraph Enforcement["Enforcement path – FMC"]
    FMConn["FMC Connector<br/>HTTPS REST API :443"]
    FMC["Secure Firewall Management Center"]
    ACP["Access Control Policy<br/>Dynamic Objects"]
    Policy -->|"Scope ↔ ACP mapping<br/>Topology-aware rules"| FMConn
    FMConn --> FMC --> ACP
    ACP -->|"auto-deploy"| FTD
end

Agentless -->|"traffic through FW"| FTD
Agent -->|"optional: rules also pushed to ACP"| Policy

```

Key idea: Visibility and enforcement use **different connectors**. NSEL tells CSW what flows exist; the FMC connector pushes **L3/L4 segmentation policy** to FTD when you enforce a workspace.

2. Data flow — visibility (NSEL)

```

sequenceDiagram
    participant W as Workload traffic
    participant FW as Secure Firewall FTD/ASA
    participant IC as Secure Firewall Connector
    participant CSW as Secure Workload

    W->>FW: Application flow
    FW->>FW: Stateful inspection
    FW->>IC: NSEL record (create/update/deny/teardown)
    Note over IC: Decapsulate NetFlow v9<br/>Forward + reverse
    flows<br/>NAT flow-stitching
    IC->>CSW: Flow observations (agentless)
    CSW->>CSW: ADM · labels · policy modeling

```

NSEL capability	Customer value
Stateful events	See permits and denies — not just sampled NetFlow
Bidirectional flows	Connector exports forward + reverse observations
Flow stitching	End-to-end visibility when NAT is in path
Agentless discovery	No agent on legacy OS, appliances, or restricted hosts

Scale (on-prem Ingest, per Cisco white paper):

Metric	Limit
Per Secure Firewall connector	Up to 45,000 fps
Per Ingest appliance (total)	Up to 135,000 fps
Connectors per Ingest appliance	1 Secure Firewall connector
Connectors per tenant	10

3. Data flow — enforcement (FMC)

sequenceDiagram

```

participant Op as Operator
participant CSW as Secure Workload
participant FMC as FMC / cdFMC
participant FTD as FTD cluster

```

```
Op->>CSW: Enable enforcement on workspace
```

```
CSW->>CSW: Resolve scope inventory → IP sets
```

```
CSW->>FMC: Create/update Dynamic Objects (WorkloadObj_*)
```

```
CSW->>FMC: Push ACP rules (Workload_* · Workload_golden_* ·
```

```
Workload_ca_*)
```

```
FMC->>FTD: Deploy policy (no manual redeploy for CSW changes)
```

Note over FTD: Topology awareness – rules only on firewalls in traffic path

FMC rule prefixes (do not edit in FMC)

Prefix	Purpose	ACP section
Workload_golden__	Allow CSW ↔ agents behind the firewall	Mandatory
Workload__	Segmentation rules from enforced workspaces	Mandatory / Default
Workload_ca__	Catch-all rules (if Use Secure Workload Catch-All enabled)	Default
WorkloadObj__	Dynamic objects for scope/inventory membership	Objects

CSW **overwrites** changes to rules using these prefixes on the next push. Independent FMC rules are preserved in **Merge** mode if they do not use these prefixes.

4. Scope ↔ ACP mapping models

flowchart LR

```

subgraph Single["Single application"]
    CS1["Child / leaf scope"]

```

```

    ACP1["One ACP"]
    CS1 -->|"1:1 mapping"| ACP1
end

subgraph Multi["Multiple applications"]
    Parent["Parent scope"]
    C1["Child scope A"]
    C2["Child scope B"]
    ACP2["One ACP"]
    Parent --> C1
    Parent --> C2
    Parent -->|"hierarchical policy"| ACP2
end

```

Mapping	When to use	What CSW pushes
Child scope → ACP	One application protected by this firewall pair	Policies for that app + parent guardrails
Parent scope → ACP	Multiple apps behind same firewall estate	Parent + eligible child scope policies

Constraint: One ACP maps to **one** scope only (1:1). Plan ACP boundaries accordingly.

5. ACP mapping options

Setting	Merge mode	Override mode
Existing FMC rules	Kept; CSW rules inserted top or bottom	Replaced by CSW rules
Rule priority	Configurable: above/below Mandatory and Default sections	N/A
Catch-all	Optional: CSW catch-all or FMC default action	Same
Dual management	Yes — network team keeps north-south / baseline rules	CSW-only in mapped sections

Policy type placement:

CSW policy type	FMC ACP section
Absolute policies	Mandatory category
Default policies	Default category

6. Firewall insertion options (on-premises)

Choose insertion based on **protection level** (typically subnet / zone boundary) and segmentation granularity.

flowchart TB

```
    subgraph L2["Layer 2 – Transparent / bump-in-wire"]
        L2W["Fine-grained · intra + inter-subnet<br/>Full NSEL visibility<br/>Legacy OS / localized workloads"]
    end

    subgraph L3["Layer 3 – Routed / gateway"]
        L3W["Faster time-to-segment · inter-subnet only<br/>Partial NSEL (no intra-subnet)<br/>Dev / non-prod zones"]
    end

    subgraph ACI["ACI + Service Graph"]
        ACIW["Intra + inter EPG/ESG<br/>PBR redirect or go-to/through<br/>CSW + FMC + ACI policies"]
    end
```

Insertion	Segmentation	NSEL visibility	Best for
L2 Transparent	Intra- + inter-subnet	Full (intra + inter)	Fine-grained, legacy OS, localized apps
L3 Routed	Inter-subnet only	Inter-subnet only	Quick zone segmentation, distributed workloads
ACI Service Graph (PBR)	Intra + inter EPG/ESG	Full with FW in path	ACI fabric + FTD service graph
ACI Go-To / Go-Through	Inter or intra+inter	Varies by mode	North-south or strict in-path designs

7. Cloud insertion options (summary)

Platform	Pattern	East-west protection	Visibility
AWS	Centralized hub VPC FTD	Inter-VPC / inter-subnet	VPC flow logs + NSEL
AWS	Distributed per-VPC FTD	Intra-VPC inter-subnet	VPC flow logs + NSEL
Azure	Hub VNet FTD + UDR	Intra- + inter-VNet	NSG flow logs + NSEL

Platform	Pattern	East-west protection	Visibility
GCP	Hub VPC FTD	Inter-VPC	VPC flow logs + NSEL

All cloud patterns support **FMC dual management**: CSW owns east-west microsegmentation; FMC owns north-south ingress/egress.

8. Extended use cases

Virtual Patch (agent + FMC)

flowchart LR

```

Agent["CSW agent on workload"] -->|"CVE inventory"| CSW
CSW -->|"FMC Connector Virtual Patch tab"| FMC
FMC -->|"Third-Party Vulnerabilities + Snort mapping"| IPS["IPS
policy / Cisco Recommended Rules"]
IPS --> FTD

```

- Requires **agents** on workloads where virtual patch applies.
- Workloads must use an **FTD address as default gateway** for traffic to hit IPS.
- FMC 7.2+ for virtual patch; CSW publishes CVEs → FMC → fine-tuned IPS rules.

Rapid Threat Containment

flowchart LR

```

Anomaly["Malware / IPS / correlation event"] --> FMC
FMC -->|"Remediation Module API"| CSW
CSW -->|"Quarantine label / guardrail policy"| Enforce["Host agent +
FTD enforce"]

```

- FMC **Remediation Module** sends affected IPs to CSW.
 - CSW **guardrail policies** (label-based) quarantine workloads across enforcement points.
-

9. Connectivity requirements

Path	Protocol	Port	Notes
FTD/ASA → Ingest	UDP (NSEL)	4729 (default)	Firewall must reach Ingest slot IP
CSW → FMC	HTTPS	443	REST API; admin credentials
SaaS CSW ↔ on-prem	Secure Connector	tunneled	Required for SaaS + on-prem FMC/Ingest
FMC HA	Include standby	443	Auto-failover to

Path	Protocol	Port	Notes
.	FMC IP		new active FMC

10. Supported versions (reference)

Capability	CSW	FMC / FTD
Segmentation (ACP mapping, topology)	3.9.1.1+	7.0.x+ (dynamic objects: 7.0.1+)
Virtual Patch	3.8.1.1+	7.2+
Simplified segmentation workflow	3.9.1.1+	7.2 recommended
Engineering qualified (white paper)	—	7.0.1, 7.2.5

Platforms: FTD devices **managed by FMC only** (not standalone ASA for enforcement path). FTD **Transparent** and **Routed** modes supported.

Official references

Document	URL
Deep dive white paper	secure.cisco.com/secure-workload/docs/secure-workload-whitepaper
FMC integration guide	cisco-secure-workload-and-fmc-integration-guide.html
Marketing white paper	sec-workload-firewall-wp.html